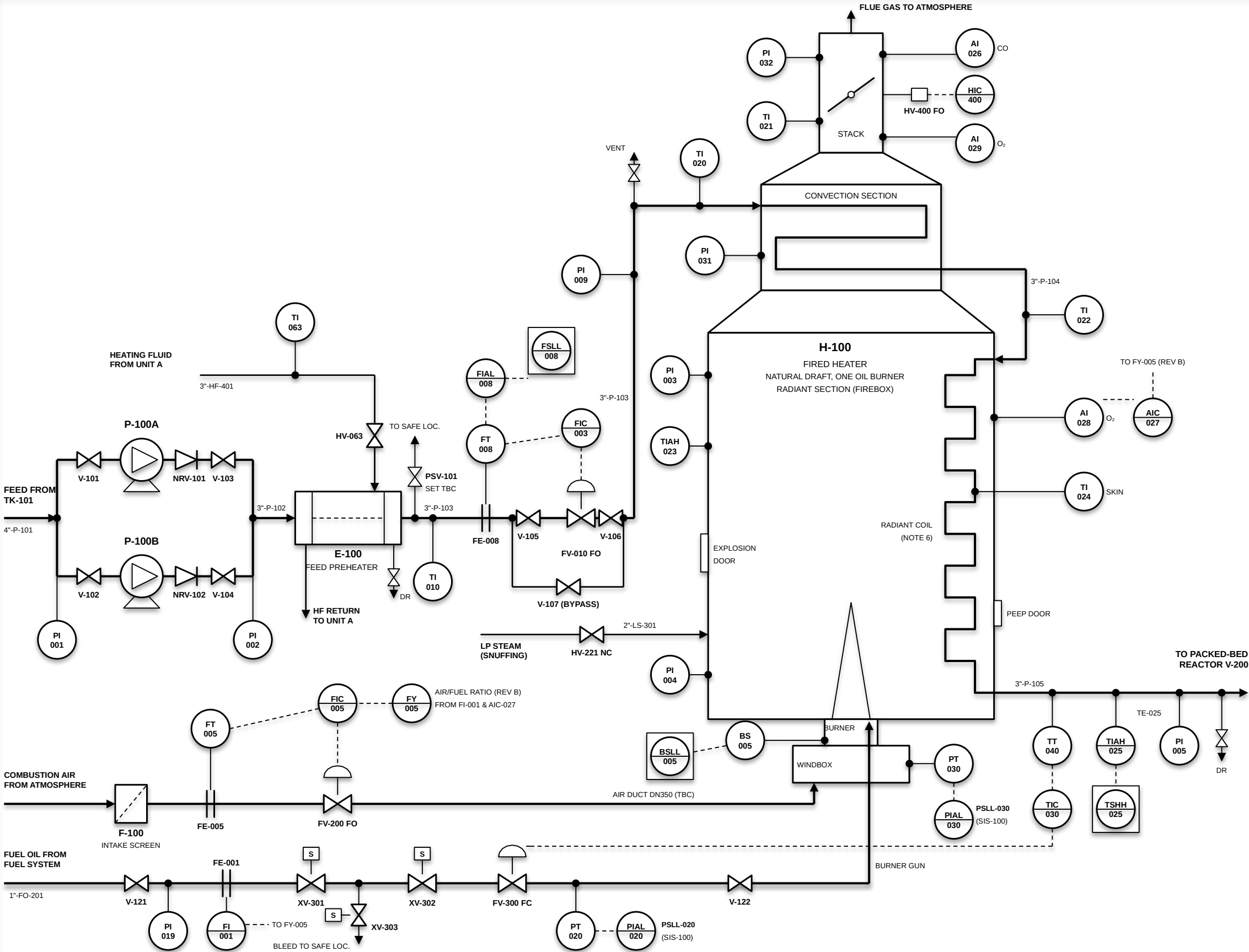




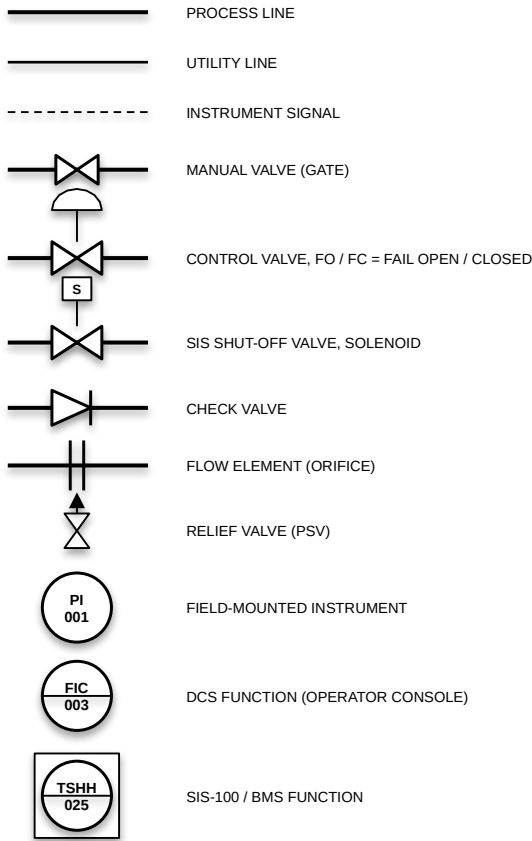
GENERAL NOTES:

- DESIGN BASIS: NATURAL DRAFT FIRED HEATER VER 1 OTS MODEL, DESIGN CASE. PRESSURES kPa g, DRAFT mmH₂O.
FEED 6.8 L/s = 19 584 kg/h (0.80 kg/L): 40 → 180 (E-100) → 233 (CROSSOVER) → 350 °C; PUMP DISCH. = 1030, OUTLET = 545.
FUEL 243 kg/h, LHV 42 MJ/kg; FIRED 2.84 MW, ABSORBED 2.31 MW (81.5 %). AIR 3 892 kg/h (11 % EXCESS).
FLUE GAS 4 380 kg/h AT 400 °C (INCL. ≈ 245 kg/h TRAMP AIR). DRAFT: ARCH -2.5, FLOOR -6.0, WINDBOX 5.25.
- LINE SIZES PRELIMINARY FROM VELOCITY (LIQUID 1.5-2.5 m/s, AIR DUCT 10 m/s) - TBC BY HYDRAULIC CALCULATION.
- PRESSURE INSTRUMENTS C/W ROOT AND TEST VALVES (TYP., NOT SHOWN). PUMP CASING VENTS/DRAINS TYP., NOT SHOWN.
- XV-301 / XV-302 / XV-303 FORM A DOUBLE BLOCK AND BLEED, OPERATED BY SIS-100 (BMS); SIGNALS NOT SHOWN.
- FV-010 FAILS OPEN SO THE COIL KEEPS FLOW ON AIR OR SIGNAL FAILURE; FV-300 FAILS CLOSED; FV-200 & HV-400 FAIL OPEN.
- RADIANT AND CONVECTION COILS SHOWN SCHEMATICALLY; TUBE COUNT AND PASSES PER DESIGN TAB (SINGLE PASS).
- FT-008 SERVES FIC-003, FIAL-008 AND FSL-008 (HAZOP GAP: ADD FT-008B FOR THE TRIP - HOLD).

NOTES (CONT.):

- TIC-030 MOVES FUEL ONLY. REV A: FIC-005 ON OWN SET POINT. REV B OPTION: FY-005 AIR/FUEL RATIO ON FI-001 + AIC-027 O₂ TRIM.
- PSV-101 PROTECTS E-100 FEED SIDE (BLOCKED-IN WITH HOT FLUID, TUBE RUPTURE). SET PRESSURE AND SIZE TBC.
- PI-019 = FUEL HEADER (≈ 1940 kPa g). PT-020 = GUN PRESSURE (≈ 600 kPa g), DOWNSTREAM OF FV-300.
- BS-005 UV FLAME SCANNER ON THE BURNER; LIGHT-OFF AND PURGE PER BMS SEQUENCE (LEGEND PANEL).
- HV-221 SNUFFING STEAM, NORMALLY CLOSED; OPEN ONLY FOR A FIREBOX FIRE. EXPLOSION DOOR ON RADIANT WALL.
- LOW-POINT DRAIN ON 3"-P-105 AND HIGH-POINT VENT AT COIL INLET FOR HYDROTEST AND DECKING.
- ALARM SET POINTS: SEE WORKBOOK (ALARM TABLE); TRIP SET POINTS: INTERLOCK SCHEDULE.

LEGEND



SERVICE CODES: P PROCESS HYDROCARBON · FO FUEL OIL
HF HEATING FLUID · LS LP STEAM · DR DRAIN
LINE NO. = SIZE - SERVICE - NUMBER

INTERLOCK SCHEDULE · SIS-100 / BMS

TRIP	CAUSE	SETTING	DELAY
FSL-008	LOW FEED FLOW	< 3.4 L/s	5 s
TSHH-025	HIGH OUTLET TEMP.	> 375 °C	5 s
PSLL-020	LOW FUEL PAT GUN*	< 40 kPa	3 s
PSLL-030	LOSS OF WINDBOX DRAFT*	< 0.5 mmH ₂ O	3 s
BSLL-005	FLAME FAILURE	NO FLAME	2 s
HS-001	ESD PUSHBUTTON	MANUAL	0 s

* ARMED WHEN FIRING.

ANY TRIP: CLOSE XV-301 & XV-302, OPEN XV-303 BLEED,
CLOSE FV-300, OPEN FV-200 & HV-400 (POST-PURGE).
RESET ONLY FROM BMS AFTER 300 s PURGE; LIGHT-OFF
AT LOW FIRE (TIC-030 MAN 10-20 %), 10 s TRIAL.

REV	DESCRIPTION	DATE
B	RATIO FY-005 + O ₂ TRIM AIC-027 (OPTION)	--
C	OTS SCREEN DRAWING (CURRENT, USED IN DECKS)	--
D	ALTERNATIVE VERSION, EPC STANDARD	29-SEP-26

NATURAL DRAFT FIRED HEATER VER 1 · OTS			
P&ID — H-100 · ALTERNATIVE VERSION			
DRAWN: APINAN SOOTTITANTAWAT	CHK'D: --	APP'D: --	DATE: 29-SEP-26
DWG NO. NDFH-V1-PID-001	REV D	SCALE N.T.S.	SHEET 1 OF 1